

Dimensionality Reduction Methods for Brain Imaging Data Analysis

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The past century has witnessed the grand success of brain imaging technologies, such as electroencephalography and magnetic resonance imaging, in probing cognitive states and pathological brain dynamics for neuroscience research and neurology practices. Human brain is “the most complex object in the universe,” and brain imaging data (*BID*) are routinely of multiple/many attributes and highly non-stationary. These are determined by the nature of *BID* as the recordings of the evolving processes of the brain(s) under examination in various views. Driven by the increasingly high demands for precision, efficiency, and reliability in neuroscience and engineering tasks, dimensionality reduction has become a priority issue in *BID* analysis to handle the notoriously high dimensionality and large scale of big *BID* sets as well as the enormously complicated interdependencies among data elements. This has become particularly urgent and challenging in this big data era.

Dimensionality reduction theories and methods manifest unrivaled potential in revealing key insights to *BID* via offering the low-dimensional/tiny representations/features, which may preserve critical characterizations of massive neuronal activities and brain functional and/or malfunctioning states of interest. This study surveys the most salient work along this direction conforming to a 3-dimensional taxonomy with respect to (1) the *scale* of *BID*, of which the design with this consideration is important for the potential applications; (2) the *order* of *BID*, in which a higher order denotes more *BID* attributes manipulatable by the method; and (3) *linearity*, in which the method’s degree of linearity largely determines the “fidelity” in *BID* exploration. This study defines criteria for qualitative evaluations of these works in terms of effectiveness, interpretability, efficiency, and scalability. The classifications and evaluations based on the taxonomy provide comprehensive guides to (1) how existing research and development efforts are distributed and (2) their performance, features, and potential in influential applications especially when involving big data. In the end, this study crystallizes the open technical issues and proposes research challenges that must be solved to enable further researches in this area of great potential.

CCS Concepts: • **Computing methodologies** → **Knowledge representation and reasoning**; **Parallel algorithms**; • **Applied computing** → **Bioinformatics**

Additional Key Words and Phrases: Brain imaging data, big data, dimensionality reduction, factorization, statistical tensor analysis, feature learning

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1 INTRODUCTION

The past century has witnessed the grand successes of brain imaging technologies in probing cognitive states and pathological brain dynamics for neuroscience research and neurology practices, with most commonly known examples including **electroencephalography (EEG)**, **magnetoencephalography (MEG)**, and **magnetic resonance imaging (MRI)**. Human brain is “the most complex object in the universe,” and brain imaging data (*BID*¹) are routinely of multiple/many attributes and highly non-stationary. These intrinsic features are determined by the nature of *BID* as the recordings of the evolving processes of the brain(s) under examination in various views, that typically manifest in (1) a *temporal course* and *frequency spectrum* directly from time-varying neural activities, (2) the *synchronization pattern* between brain regions, (3) and *population difference* (similarity or dissimilarity) caused by subject-specific physiological or psychological states, making *BID* analysis largely different from general image analysis as depicted in Figure 1. Determination of the principles and standards for sophisticated *BID* processing and analysis is critical to adapt to the tremendous complexity [29, 91].

Driven by the increasingly high demands for precision, efficiency, and reliability in neuroscience and engineering tasks, dimensionality reduction has become a priority issue in *BID* analysis to handle the notoriously high dimensionality and large scale of massive *BID* sets as well as the enormously complicated interdependencies among data elements. This has become particularly urgent and challenging in this big data era. Dimensionality reduction theories and methods manifest unrivaled potential in revealing key insights to *BID* via offering the low-dimensional and often tiny representations [22, 77], and these are actually latent features (factors) preserving critical characterizations of massive neuronal activities and brain functional and/or malfunctional states of interest. The ultimate goal of dimensionality reduction is to establish a mapping relationship from the high-dimensional data space to the low-dimensional feature space [33].

The features (factors) of *BID* can be specifically characterized in various views—that is, being examined by dynamic structure, spectrum analysis, topographic distribution, and population discrimination, among others. Dimensionality reduction methods therefore play an extremely important role in exploratory analysis such as neurophysiological assessment and task-oriented applications such as **brain-computer interface (BCI)** tasks.

However, there exists a pressing need to bridge the gap between the diversity of a large amount of dimensionality reduction theories and methods and the huge diversity of *BID* and more importantly the neuro-science and engineering tasks. Grand challenges arise when facing the following critical concerns:

- (1) *Effectiveness*: How to choose the appropriate and effective method to sufficiently capture the intrinsic features of *BID* to fulfill the demanded tasks? For example, spatial filter CSP (common spatial patterns) works particularly well in **motor-imagery (MI)** BCI, where the low-dimensional features are sharply distinct from each other in different tasks. In contrast, CSP fails in P300 detection (**event-related potential (ERP)** lasting for 100 ms) for the

¹For instance, EEG/MEG signals are time series data organized in tensor streams [112, 121], whereas MRI acquisitions are image sequences organized in tensor of limited length [45].

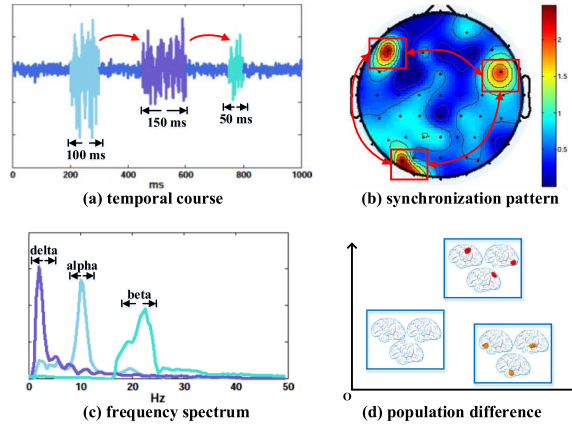


Fig. 1. Illustration of *BID*'s features.

sensitivity to noises and a poor performance of evoked area identification. Note that the former is a long-term task, whereas the latter is a short-term one.

- (2) *Interpretability*: How to obtain scientific understandings or interpretations on the overall mechanisms or those of interests regarding brain states, especially under mild conditions? Exploratory analysis is probably the ultimate goal of most *BID* applications, where interpretability is at the very core of analysis. Taking the analysis of pathological EEG with seizures, for example, factorization-based methods are far superior to low-dimensional counterparts, such as traditional time-frequency analysis, to characterize epileptic states by correlating the modes all together: frequency, temporal course, and space [22].
- (3) *Scalability*: How to solve increasingly large neuroscience and engineering problems? Conventional dimensionality reduction methods are rooted from the community of mathematics and physics with the “correctness” of extracted features as the major concern. Their designs leave large problems unattended but focus on the innovation of the fundamental mathematical theory [28]. The “Divide-and-conquer” scheme is worth consideration for reaching the goal when the problem may be partitioned, even though the time for problem solving is likely to be prohibitively long.
- (4) *Efficiency*: How to adapt to the time-critical neuroscience and engineering applications with the routinely high complexity of dimensionality reduction algorithms? For example, BCI tasks demand real-time *BID* analysis, whereas the performance bottleneck is a constant for most dimensionality reduction methods. Recent emerging big *BID* technologies have driven the trend of optimizing the process of dimensionality reduction and/or improving the efficiency with advanced computing technologies. The smart medical system based on the Internet of Things is exactly a successful case [57, 67].

How do we determine what kind of dimensionality reduction (*DimRed*) methods to choose or develop largely lie with the answers to the preceding concerns? This study defines the criteria for qualitative evaluations of existing *DimRed* theories and methods in terms of effectiveness, interpretability, efficiency, and scalability from the perspective of *BID* applications.

Vast research on *DimRed* of *BID* exists, ranging from fundamental mathematical theories to systems and applications in neuroscience and engineering practices. However, the few existing surveys focus on a simple classification of *DimRed* methods by discriminating their underlying mathematical bases and the algorithms' significances in capturing useful information from

high-dimensional data [34, 151]. How these may apply in various scenarios of *BID* analysis is largely ignored. There exists a pressing need for a comprehensive survey on them with a systematic taxonomy. In light of this, this study aims to aid researchers and developers in this young and promising area (1) to have an in-depth understanding of the various principles of *DimRed* centering on *BID* and the issues addressable by *DimRed* methods with theoretical details recapped, and (2) to well match the features of candidate *DimRed* methods with *BID*'s characteristics under specific scenarios, especially the applicability in big *BID* problems.

To this end, this survey overviews the most salient work along this direction conforming to a 3-dimensional taxonomy with respect to the following:

- (1) *The scale of BID*: The design with this consideration is important for potential applications (from conventional to optimized algorithms toward big *BID*). Clearly, the computing issues have become more and more a priority and challenging in any contemporary *BID* application in this big data era.
- (2) *The order of BID*: A higher order denotes more *BID* attributes manipulatable by the method (from low order to high order). Low-order methods are prone to a satisfied convergence but incapable of maintaining the interactions among multiple attributes of *BID*. High-order methods are superior in preserving the interdependencies among data elements in each domain but notoriously are compute intensive, such as matrix factorization versus tensor factorization.
- (3) *Linearity*: The method's degree of linearity largely determines the "fidelity" in *BID* exploration (from non-linear to linear). Linear methods excel in simplicity and interpretability but may be insufficient or require latent features of a much larger number than that of data dimensions to reveal the implicitly non-linear characteristics of *BID*. Non-linear methods can fit the non-linearity of *BID* but demand intensive computation, such as standard PCA (principal component analysis) [1] versus kernel methods based on PCA [160].

In general, *BID* exhibit intensive non-linearity in a high-dimensional data space. Note that "non-linear" and "high-order" are two intrinsic attributes of *BID* stressed in this study. In addition, analyses of *BID* often need to consider the influences of other bio-signals such as **electromyography (EMG)**, **electrooculography (EOG)**, and **electrocardiography (ECG)** [108]. Figure 2 overviews the most popular *DimRed* methods against a diversity of the dominant brain imaging technologies and the accompanying bio-signal technologies.

The main contributions of the work are listed as follows:

- (1) This survey proposes a 3-dimensional taxonomy and four evaluation criteria on dimensionality reduction methods for *BID* analysis to provide comprehensive guides about how existing research and development efforts are distributed along with corresponding pros and cons discussed (Section 3, Section 5);
- (2) This survey presents the most salient methods' features, performance, and potential in influential applications especially when processing big *BID* (Section 4). It recaps their theoretical bases and discusses the considerations in various scenarios of *BID* analysis.
- (3) This study crystallizes the open research issues and proposes challenges that must be solved to enable further researches in this area of great potential (Section 5).

2 DEFINITIONS AND NOTATIONS

In this study, scalar, vector, matrix, and N -order tensor are respectively denoted as x , $\mathbf{x}$, $\mathbf{X}$, and $\mathcal{X}$. To avoid ambiguities between "dimensionality reduction" in the general sense and "tensor (multi-way data) decomposition," the definition of "dimension" in this study emphasizes the following.

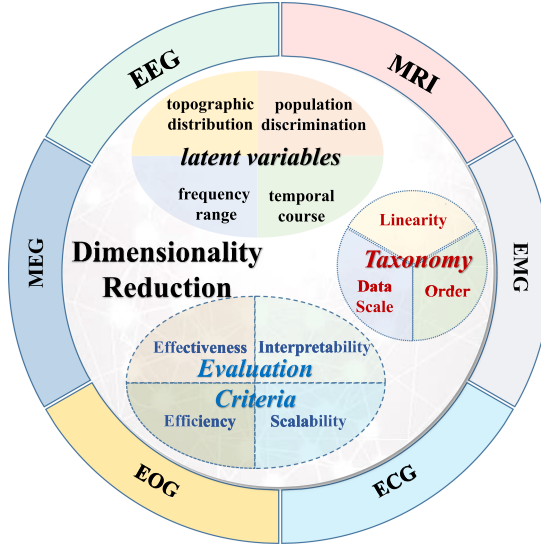


Fig. 2. Overview of *DimRed* methods for *BID* and accompanying bio-signal data.

First, for the former cases, “dimension” denotes the length of each data item—for example, $\mathbf{x} = (x_1, x_2, x_3)^T$ is 3-dimensional, commonly referred to in machine learning community. Second, for the case of tensor decomposition, “dimension” characterizes the structure of multi-way data—for example, vector (1-dimension), matrix (2-dimension), and tensor (3-dimensions or more). In this context, dimension is denoted as order/domain/mode in the rest of this survey, namely vector (1-order), matrix (2-order), and tensor (3-orders or more).

DimRed methods are generally conforming to the basic principles as stated in Definitions 1 and 2.

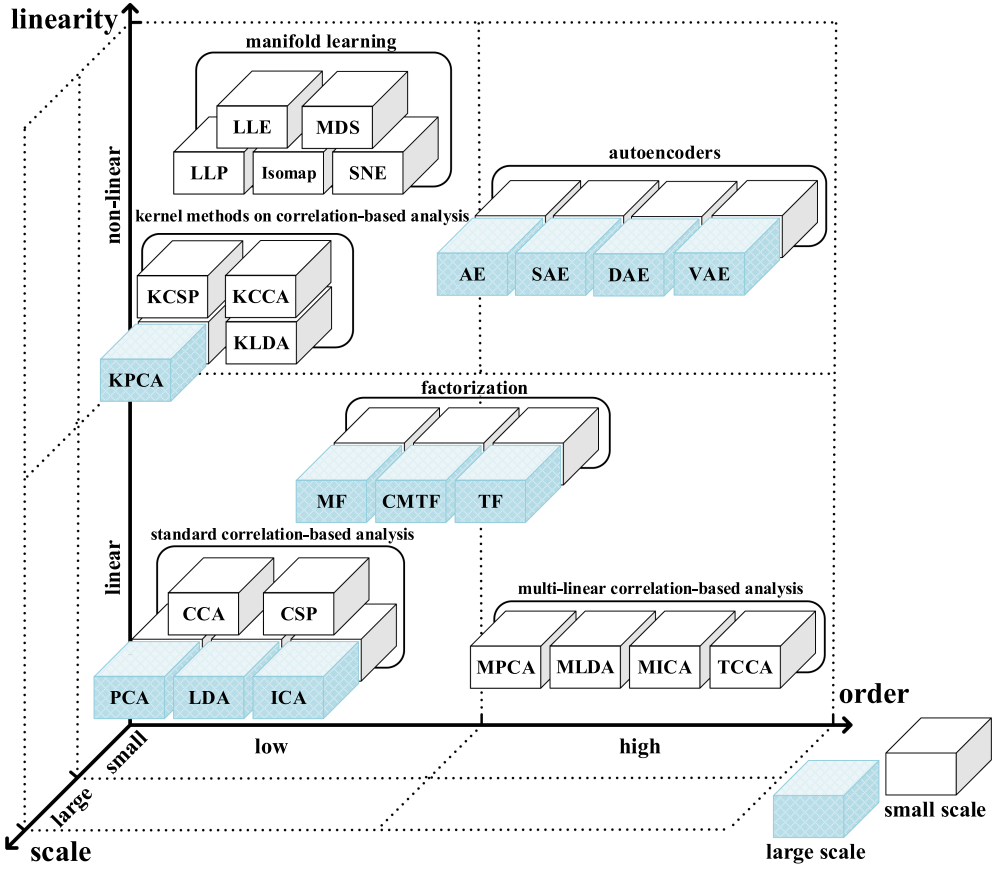
Definition 1 (Low Order). Given sets of observed variables $X = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m)^T \in \mathbb{R}^{m \times d}$ and the positive integer $r (r < d)$, latent variables $Y = (\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_m)^T \in \mathbb{R}^{m \times r}$ are obtained through a transformation $Y = f_\Theta(X)$, call Θ mapping rule, and $f_\Theta(\cdot)$ mapping function.

Definition 2 (High Order). Given sets of observed variables $\mathcal{X} = (X_1, X_2, \dots, X_m)^T \in \mathbb{R}^{m \times I_1 \times I_2 \times \dots \times I_N}$ and the positive integer sets $\{J_n\}_{n=1}^N (J_n < I_n)$, latent variables $\mathcal{Y} = (Y_1, Y_2, \dots, Y_m)^T \in \mathbb{R}^{m \times J_1 \times J_2 \times \dots \times J_N}$ are obtained through a transformation $\mathcal{Y} = f_\Theta(\mathcal{X})$, call Θ mapping rule, and $f_\Theta(\cdot)$ mapping function.

In the context of *DimRed* of *BID*, observed variables or observations denote (high-dimensional) *BID* (or some parts) under exploration, whereas latent variables are exactly low-dimensional features/representations/factors to be derived. *BID* may also be processed as canonical variables to simplify the problem in a convenient way based on the nature of the problem [51]. These terms will be referred to interchangeably except when specified.

Note that Definitions 1 and 2 define *DimRed* of data organized in a low order and a high order, respectively. The primary challenge here is to infer the mapping function $f_\Theta(\cdot)$ from the known mapping rule Θ by means of optimizing the objective function:

$$\hat{f} = \arg \min_f F(f). \quad (1)$$

Fig. 3. Classification of *DimRed* methods.

DimRed methods aim at the common goal depicted in Equation (1) but differ in the establishment of mapping rule Θ . More specifically, linear methods are explicitly defined to conform to $Y = f_{\Theta}(X) = XW$ between latent variables Y and observed variables X , where $W = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_r) \in \mathbb{R}^{m \times r}$ are sets of mapping vectors. On the contrary, non-linear methods are generally denoted as $Y = f_{\Theta}(X)$ with implicitly non-linear transformation in $f_{\Theta}(\cdot)$.

3 OVERALL PICTURE OF THE *DIMRED* METHODS

This section overviews the categories of existing *DimRed* methods, which draws an overall picture about how existing research and development efforts are distributed according to 3-dimensional taxonomy (scale, order, linearity) with qualitative evaluations provided in terms of the four criteria (effectiveness, interpretability, scalability, efficiency).

In this survey, *DimRed* methods are classified with each category corresponding to a region in the 3-dimensional space constructed by the axes of scale (small to large), order (low to high), and linearity (linear to non-linear) as shown in Figure 3. Considering the trend of big *BID* applications, each cube in Figure 3 is highlighted with its applicability to a scenario of small-scale *BID* only (blank) or large-scale *BID* too (blue).

The most salient *DimRed* methods are generally rooted from correlation-based analysis (**standard correlation-based analysis (SCA)**), kernel methods on correlation-based analysis,

and multi-linear correlation-based analysis), factorization, manifold learning, and autoencoders, which inherently differ from each other in their mathematical bases (see Section 4 for details):

(1) *Correlation-based analysis*:

- SCA methods (low order, linear, and small/large scale) aim to find sets of the most representative projection vectors to establish the linear relation between latent variables and observed variables (Section 4.1). Typical examples include PCA, LDA (linear discriminant analysis) [14], ICA (independent component analysis) [134], CSP (common spatial pattern) [16], and CCA (canonical correlation analysis) [51]. These derive latent variables Y from observed variables X through a linear transformation $Y = XW$ as defined in Section 2, usually handling data organized in low order. To cater to the need of big *BID* applications, scalable and parallel extensions to PCA, LDA, and ICA have been developed, including fast PCA [128], incremental LDA [118], and CUDAICA [126].

- Kernel methods on correlation-based analysis (low order, non-linear, small/large scale) center on a class of algorithms to improve a linear model to a non-linear one with “kernel tricks” (Section 4.3). SCA relies on the inner product of pairs of sample points as $\langle x_i, x_j \rangle$ in its core algorithm, whereas kernel methods employ an alternate kernel function, such as multinomial function and Gaussian function, to transform $\langle x_i, x_j \rangle$ into $\langle \phi(x_i), \phi(x_j) \rangle$ with non-linear mapping function $\phi(\cdot)$. Non-linear *BID* analysis is then enabled with the non-linear function implicitly embedded. Typical examples consist of KPCA (kernel PCA [144]), KLDA (kernel LDA [98]), KCSP (kernel CSP [35]), and KCCA (kernel CCA [17]). Kernel methods suffer from high memory and runtime overhead when processing massive *BID*. To address the issues, KPCA has been extended for incremental and dynamic analysis, such as dynamic KPCA [25] and fast KPCA [49].

- Multi-linear correlation-based analysis methods (high order, linear, small scale) extend the standard counterparts with multi-linear algebra, which directly handle high-order data (tensor) instead of having to matricize the data as low-order counterparts do (Section 4.2). These methods determine the projection bases $\{W_n\}_{n=1}^N$ in multiple domains to capture the most informative representations presented in the original data, where latent variables $\mathcal{Y}$ are derived through the transformation $\mathcal{Y} = f_{\Theta}(X) = X \times_1 W_1 \times_2 W_2 \times \cdots \times_N W_N$ as defined in Definition 2 ($\times_n$ is a tensor-matrix product). Typical examples along this direction include MPCA (multi-linear PCA) [102], MLDA (multi-linear discriminant analysis) [158], MICA (multi-linear ICA) [147], and TCCA (tensor CCA) [71]. These are currently limited to offline and static analysis of small-scale *BID*.

(2) *Factorization*: Factorization methods (low/high order, linear, small/large scale) decompose observed data into multi-domain/mode/way latent components through low-rank approximation(s) [28] (Section 4.1, Section 4.2). This survey focuses on three types of factorization models: (1) basically, MF [59] derives latent components (coefficient matrix U and basis matrix V) from observed data X in a linear manner as $X = UV^T$, where the input data X is organized as a matrix (low order); (ii) TF [28] extends MF to factorize data of higher dimensions, which directly derives the multi-domain latent components of a tensor X ; and (3) **coupled matrix and tensor factorization (CMTF)** [4] is based on MF and TF. CMTF aims to facilitate joint *DimRed* of multi-modal data with one or more domains shared, where input data are organized as either matrix or tensor. Numerous attempts have been already made to adapt factorization methods to big *BID* on the basis of the preceding factorization models, such as online NMF [106], IraNTF [168], and SCouT [62].

(3) *Manifold learning*: Manifold learning (low order, non-linear, small scale) preserves the local/global structure of high-dimensional data (observations) and constructs low-dimensional manifolds from data points in the original data space (Section 4.3). Typically, MDS (multi-dimensional scaling) applies to keeping the distance between each pair of data points unchanged in the low-dimensional manifold and high-dimensional data space [63]. Without the assumption of global linearity as required in correlation-based analysis, manifold learning assumes a weaker local linearity to well fit the non-linear data with data points organized as vectors (low order). Other instances

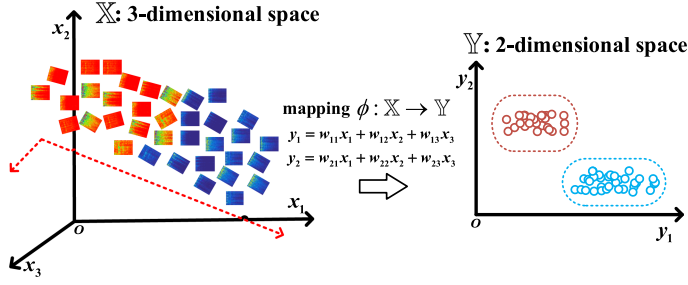


Fig. 4. Illustration of linear & low-order *DimRed* methods. Observations (EEG) in 3-dimensional space $\mathbb{X}$ are transformed into those in 2-dimensional space $\mathbb{Y}$ by mapping ϕ : $y_1 = w_{11}x_1 + w_{12}x_2 + w_{13}x_3$, $y_2 = w_{21}x_1 + w_{22}x_2 + w_{23}x_3$.

have also achieved grand success, including Isomap (isometric feature mapping) [139], LLE (locally linear embedding) [127], LPP (locality preserving projections) [52], and SNE (stochastic neighbor embedding) [55]. Two pending issues remain, however: time-consuming graph manipulation and limited capacity in incremental learning when handling massive data [138].

(4) *Autoencoders*: Autoencoders (high order, non-linear, small/large scale) are unsupervised neural networks. The designs emphasize how to compress/encode data and reconstruct/decode data back from the encoded representations (Section 4.4). Autoencoders are data driven by nature, where the compressed representations can be learned from the original data possibly of a high dimension. Raw **autoencoder (AE)** [56] achieves this goal by minimizing the reconstruction error. Successful attempts have been also made to enhance AE's learning capacities, such as **stacked autoencoder (SAE)** [130], **denoising autoencoder (DAE)** [150], and **variational autoencoder (VAE)** [72]. These methods support non-linear transformations via non-linear activation functions, such as tanh-function and sigmoid-function, and manipulate data in a high-order manner, such as 3D convolution. Autoencoders manifest a build-in high parallelism, and their outstanding performances rely on the training process of large-scale data. These features make them suitable for large-scale *BID* when sufficient computing resources are available.

For researchers and developers in this area, Figure 3 provides a clear guideline for them to determine the initial method(s) to fulfill their specific tasks, especially when handling big *BID* problems.

4 SURVEY ON THE *DIMRED* METHODS FOR *BID* ANALYSIS

This section surveys the *DimRed* methods on basis of the overall picture presented in Section 3, which is organized following two dimensions of the taxonomy: linearity and order. For each surveyed method, its fundamental mathematical theory is described, followed by its *BID* applications (in “small data” era by default) with the capability of handling large-scale *BID* evaluated when applicable, i.e., scale dimension of the taxonomy. In big data era, existing static methods cannot directly apply as the computing issues have not been considered at the first, i.e., scalability and efficiency.

4.1 Methods: Linear and Low Order

Linear and low-order. DimRed methods map observed variables to latent variables in a linear manner with a low-order model suitable for data with low-order organization (depicted in Figure 4). Methods in this category can be rooted from SCA and factorization.

4.1.1 Mathematical Models. SCA methods explore the correlations between variables or within variables to find sets of the most representative projection vectors and thus establish the linear relationship between latent variables and observed variables. PCA, LDA, ICA, CSP, and CCA are the most salient instances in this category.

- PCA is probably one of the most typical algorithm for *DimRed* of *BID*. PCA employs the orthogonal transformation that converts a set of observed variables into a set of variables with linear uncorrelation (principal component) in an unsupervised manner. As a result, the greatest variance of transformed data lies on the first coordinate, the second greatest variance on the second coordinate, and so on [1]. The goal of PCA is to maximize the variance of low-dimensional representations as described in Appendix A.1. Then Equation (A.1.1) can be simply solved by EVD (eigenvalue decomposition) or SVD (singular value decomposition). Efforts have also been made to improve PCA's robustness and stability, such as sparse PCA [65].

- LDA, also known as Fisher's linear discriminant, always manifests a counterpart to PCA in a supervised learning manner. LDA explores a linear transformation to separate two or more classes of data points by making divergence between classes maximized and divergence within classes minimized in the subspace. The optimization objective of LDA is available in Appendix A.2. The solution to Equation (A.2.1) can be available by addressing the generalized eigenvalue problem in the work of Bickel and Levina [14]. When handling multiple classes, LDA redefines the inter-class scatter matrix as $S_b = \frac{1}{C} \sum_{c=1}^C (\mu_c - \mu)(\mu_c - \mu)^T$ rather than $S_b = (\mu_0 - \mu_1)(\mu_0 - \mu_1)^T$ in Equation (A.2.1), where μ is the mean of classes. Then the objective is optimized as before to fulfill a multi-class *DimRed*.

- Similar to LDA, CSP also adopts a supervised fashion that makes the differences of variance between two or more classes of time series as maximum as possible. The goal of CSP can be obtained in Appendix A.3. Analogously, Equation (A.3.1) is transformed into an eigenvalue decomposition problem to be easily solved. CSP yields a high variance for observations in class-0 and a low variance for observations in class-1, and vice versa, and thus can significantly enhance differences between various classes [16]. In this sense, CSP is particularly suitable for BCI applications. Later, variants to multi-class paradigms have been proposed by Gross-Wentrup and Buss [46] to facilitate multi-class BCI tasks.

- ICA is the natural choice for applications involving blind source separation. ICA treats multivariate signals as mixtures of independent non-Gaussian sources and attempts to find the linear transformation among them. Given observations $X \in \mathbb{R}^{m \times d}$ and the number of sources $r (r < d)$, the general ICA model can be expressed as $S = XW$, where $S \in \mathbb{R}^{m \times r}$, $W \in \mathbb{R}^{d \times r}$ respectively indicate the sources and unmixing matrix. For obtaining the solution of ICA (W, S), two well-known methods are highlighted (Infomax ICA [11], FastICA [134]), which aim to maximize the joint entropy of sources S depicted as $H(S) = \frac{1}{m} \ln p_s(XW) + \ln |W|$, where $p_s = 1 - \tanh(s)^2$.

- CCA mainly explores the relations between two sets of variables that always find the linear combinations of them to achieve a joint *DimRed* [51]. More specifically, CCA aims to maximize the correlation coefficient between two sets of low-dimensional variables with optimization objective described in Appendix A.4. In the same way, Equation (A.4.1) can be solved by singular value decomposition.

Factorization methods seek to decompose observed data into multi-domain latent components with low-rank approximation [28]. Unlike correlation-based analysis methods that rely on strict assumptions, such as statistical independence between latent sources in ICA, factorization methods consider that the observations are regulated by small number of latent factors, where factors are artificially introduced causes to explain the behavior of original data.

- MF is the only factorization method in this category, which decomposes a data matrix into the coefficient matrix and basis matrix. Namely, given observations $X \in \mathbb{R}^{m \times d}$ and the number of

Table 1. Summary of Linear and Low-Order Methods from the Perspective of the Objectives for Optimization

Method	Learning Pattern	Goal	Objective Function
PCA	Unsupervised	Maximize the variance	$\arg \max_{W^T W = I_r} \text{trace}(W^T S W)$
LDA	Supervised	Maximize inter-class divergence and minimize intra-class divergence	$\arg \max_{W \in \mathbb{R}^{d \times r}} \frac{\text{trace}(W^T S_b W)}{\text{trace}(W^T S_w W)}$
CSP	Supervised	Maximize the variance between different classes	$\arg \max_{W \in \mathbb{R}^{n \times r}} \frac{\text{trace}(W^T R_0 W)}{\text{trace}(W^T R_1 W)}$
ICA	Unsupervised	Maximize joint entropy of sources	$\arg \max_{W \in \mathbb{R}^{n \times r}} \frac{1}{m} \ln p_s(XW) + \ln W $
CCA	Unsupervised	Maximize correlation coefficient	$\arg \max_{\substack{U^T \text{cov}(X, X)U = I_r \\ V^T \text{cov}(Y, Y)V = I_r}} \text{trace}(U^T \text{cov}(X, Y)V)$
MF	Unsupervised	Minimize mean square error	$\arg \min_{U \in \mathbb{R}^{m \times r}, V \in \mathbb{R}^{d \times r}} \ X - UV^T\ _F^2$

latent factors r , MF aims at finding matrices $U \in \mathbb{R}^{m \times r}$ and $V \in \mathbb{R}^{d \times r}$ to suffice that $X = UV^T$. To obtain factor matrices U and V , the commonly used method is to minimize the mean square error ($\|X - UV^T\|_F^2$). Then the solution can be easily derived through gradient descent.

For applications in practice, nonnegative constraint always applies to reinforce the derivation of nonnegative factors for interpretability purpose (**nonnegative MF (NMF)**), where elements in U, V are equal or greater than 0. In addition, other constraints may also apply: (1) sparseness constraint is added to learn parts-based representations [59], and (2) orthogonality constraint can be imposed onto either U or V to minimize goodness-of-fit [24]. A robust variant of NMF embeds the L_{21} norm to handle noises and outliers, which is a constant in applications involving noisy *BID* such as EEG [73].

Table 1 offers a comprehensive summary of these methods from the perspective of the objectives for optimization.

4.1.2 Typical Applications. SCA and MF have been widely applied in *BID* analysis benefiting from the simplicity of solving their models.

- PCA embodies superiority in feature extraction particularly suitable for epileptic EEG analysis [5, 133] and artifact removal [19, 140]. Acharya et al. [5] and Siuly and Li [133] utilize PCA for *DimRed* of epileptic EEG and obtain few significant eigenvalues via an ANOVA test, thus enabling abnormality detection of an acceptable accuracy at a relatively low computational cost. PCA is also applied in suppressing artifact components, such as ocular artifacts, in TMS-EEG (TMS: transcranial magnetic stimulation) and ERP-EEG, to easily reveal the components of evoked potential and to increase the number of useful trials [19, 140]. PCA may significantly assist classification tasks such as emotion recognition [64] and schizophrenia discrimination [18]. Jirayucharoensak et al. [64] alleviate the overfitting problem in the emotion recognition task due to the intensive non-stationarity of EEG, where PCA with adaptation to covariate shift extracts the key components from the *PSD* (power spectral densities of EEG); the accuracy of emotion recognition increases by 5.55%.

- LDA mainly applies to labeled *BID* as classifiers to directly fulfill the classification tasks along with discriminant feature construction, such as seizure detection [170] and MI/ERP classification [161, 165]. Epileptic EEG first goes through wavelet decomposition, then LDA processes lacunarity and fluctuation indices of wavelet coefficients to detect epileptic seizures [170]; the method achieves high performance with sensitivity of 96.25% and a false detection rate of 0.13

per hour at a mean delay of 13.8 seconds. Note that LDA assumes Gaussian distribution of data with equal covariance matrices for concerned classes. The assumption is often too strict for MI classification. Zhang et al. [161] propose a z-score LDA to handle the issue of heteroscedastic class distributions in MI EEG. The z-score LDA treats the mean and standard deviation of the projected data as the z-score and adaptively adjusts the decision boundary to fit the heteroscedastic distribution, and the classification accuracy has been enhanced in contrast to the conventional LDA.

Furthermore, the performance of LDA is largely affected by the bias-variance trade-off, especially under the circumstance of insufficient data. The aggregation of sparse LDA has then been introduced using L_1 -regularized least squares regression, where multiple sparse discriminant vectors are learned by exploiting the equivalence between LDA and the least squares regression; with promising performance in ERP classification, the method holds potential BCI applications with small data in practice [165].

- As introduced previously, CSP prevails in spatial filtering, especially for EEG-based BCI tasks, which constructs optimal features to discriminate EEG measures of different classes. CSP analyzes the spatial pattern of imagined hand and foot movements and significant channels can be selected by searching the maximums of spatial pattern vectors, thus achieving high classification accuracy up to 93.45% and 91.88% [153].

Note that CSP's performance largely depends on the appropriate selection of a subject-specific frequency range and spatial distribution. Numerous attempts have been made to extend CSP to address this issue, such as SBCSP, FBCSP, DFBCSP, and PCSP. Novi et al. [113] propose SBCSP (sub-band CSP) to overcome the problem of the onerous fine-tuning process to adapt to the changeable rhythmic patterns from different subjects [113]. SBCSP decomposes EEG into sub-bands using a Gabor filter bank to extract the discriminative features along each sub-band, which enhances the robustness and consistency of the solution to BCI tasks. FBCSP (filter bank CSP) extends SBCSP by considering the relations between sub-bands using mutual information to choose a small subset of effective spatial filters [8]. FBCSP is capable of learning subject-specific patterns to ensure a relatively high classification accuracy. Thomas et al. [141] develop DFBCSP (discriminative filter bank CSP) to enhance the generalization ability of FBCSP using the Fisher ratio of spectral estimation on channels C3 and C4 to select discriminative filter banks [141]; the method supports a subject-specific learning pattern and holds potential in online BCI application. More variants are successively proposed to tackle model overfitting caused by noises or insufficiency of EEG data, such as RCSP (regularized CSP) [101] and PCSP (probabilistic CSP) [157].

- ICA can decompose signals into source components and artifactual components without supervision, which achieves grand success in artifact removal of *BID* [61, 149]. For instance, Vigário et al. [149] employ ICA to extract different types of artifacts (activities of eyes and muscles) from EEG/MEG data evoked by sensory stimuli, and thus responses between the ipsilateral region and contralateral region of the brain can be explicitly discriminated by removing those artifacts. ICA with uncorrelation assumption on source signals has also separated sources from noise-contaminated MEG data [61], which can reliably estimate the number of sources and thus remove the noises.

- CCA examines the underlying correlations of multi-group *BID* of a single or multiple modalities, mainly for artifact removal [37] and **steady-state visual evoked potential (SSVEP)** recognition [95]. CCA applies to remove muscle artifacts in EEG and motion artifacts in EEG and fNIRS (functional near-infrared spectroscopy) in a blind source separation manner, and thus the underlying source activities can be effectively strengthened [17, 37]. Lin et al. [95] exploit CCA to extract frequency features and representative channels of SSVEP for the first time with a high accuracy achieved in SSVEP recognition. In addition, a multi-set CCA is proposed to extract common features of multiple EEG datasets recorded at the same stimulus frequency [164], which

outperforms conventional CCA with more accurate characteristics of SSVEP captured. CCA manifests potential in joint analysis of multi-modal *BID*. Correa et al. [32] use CCA to simultaneously decompose ERP and **functional MRI (fMRI)** into spatial maps and the corresponding temporal evolution [32]; the method can highlight spatial activations in motor, frontal, temporal, and inferior parietal areas matching the presented paradigm; it facilitates the study of brain connectivity and localization of brain functions.

- MF represents *BID* as a weighted linear combination of latent factors with each factor highlighting some salient features. MF mainly applies in EEG feature extraction [84, 85] and fMRI analysis [6, 99]. Lee et al. [84, 85] propose kernelized NMF and group NMF to learn discriminative spectral feature and task-related common bases with both intra-subject and inter-subject variations of EEG reflected; both methods excel in classification tasks as a contrast to the ordinary counterparts. Lohmann et al. [99] utilize NMF to elicit intrinsic structure of single-trial data with interdependencies between trials across brain regions revealed when analyzing fMRI; NMF also applies to infer the latent structure of fMRI to detect differential changes in **attention deficit hyperactivity disorder (ADHD)** [6]. The latent variables may be referred to as “biomarkers” to differentiate the ADHD and healthy patients. Damon et al. [36] also employ NMF in artifact removal (ocular artifact) of EEG with satisfactory performance achieved.

4.1.3 Potential in Big *BID* Applications. Numerous attempts have been made to optimize the methods of SCA and factorization to adapt to big *BID* problems. The most salient work along this direction mainly concerns PCA, LDA, ICA, and MF.

- The first priority has been given to PCA, and the high-performance family of PCA consists of fast PCA [128], parallel PCA [7], distributed PCA [92], and scalable PCA [41]. Sharma and Paliwal [128] develop fast PCA (FastPCA) using the Gram-Schmidt orthogonalization process to find leading eigenvectors with a speedup of more than 100 times obtained versus traditional PCA. Parallel PCA (GS-PCA) extends traditional PCA to operate on a **graphics processing unit (GPU)** [7], and thus a 12-time speedup is gained with its massively parallel computing capability. Liang et al. [92] improve PCA by computing subspace to capture as much variance of the union of data points as possible in a distributed architecture (disPCA) [92], and thus PCA can be accelerated with several orders of magnitude. In addition, scalable PCA (sPCA) takes advantage of MapReduce on Spark with three key innovations: (1) to optimize large matrix operations, (2) to leverage matrix sparsity, and (3) to minimize intermediate data; it excels in runtime efficiency and storage utilization [41].

- Incremental LDA [118, 167] and parallel ICA [40, 126] significantly improve the performance of LDA and ICA. Pang et al. [118] introduce incremental LDA (ILDA) with new data progressively added to the initial eigenspace [118]; the method can effectively update discriminant eigenspace over high-throughput and massive data streams with discriminant features extracted for classification. Zhao and Yuen [167] propose an alternative incremental LDA by incorporating a generalized SVD to determine the projection matrix in full space (GSVD-ILDA), which excels in low computational complexity and high scalability. In addition, parallel ICAs have been developed to operate on CUDA (CUDA-ICA) [126] and FPGA (FPGA-ICA) [40] with multiplications of vector-matrix and matrix-matrix optimized, which demonstrates potential in processing large-volume data with a speedup of 25 $\times$.

- Optimized MF variants mainly include online MF [48, 106], large-scale MF [43, 105], and optimal NMF [47]. Conventional NMF requires the entire data to reside in the memory, whereas this is impossible for massive *BID* or streaming *BID*. Mairal et al. [106] propose online MF (online-MF) for factorization of large matrices using stochastic approximation; the method gracefully scales to datasets consisting of millions of data elements. Guan et al. [48] propose another online NMF (OR-NMF) using a quasi-martingale and buffering strategy to support incremental learning; the method

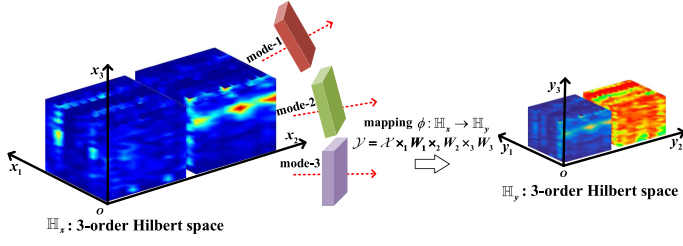


Fig. 5. Illustration of linear and high-order *DimRed* methods. The 3-order observations (EEG) in Hilbert space $\mathbb{H}_x$ are transformed into those with comparatively small scale in each domain in another Hilbert space $\mathbb{H}_y$ by mapping $\phi: \mathcal{Y} = \mathcal{X} \times_1 W_1 \times_2 W_2 \times_3 W_3$.

possesses two merits, making it particularly suitable for large-scale and streaming *BID*: (1) the time and space complexities remain constant in each step when the data scale increases, and (2) the local optimal solution can always be found. In addition, a large-scale MF algorithm has been designed to factorize dense matrices with billions of elements by exploiting the “stratified stochastic” gradient descent (DSGD-MF) [43]; the algorithm can be fully distributed, converge quickly, and scale well. Mackey et al. [105] introduce a “Divide-Factor-Combine” NMF (DFC-MF) framework with near-linear to super-linear speedups with data scale increasing: (1) to divide large-scale MF problems into small sub-problems, (2) to solve each sub-problem in parallel, and (3) to combine sub-problem solutions.

In contrast to the preceding, Guan et al. [47] focus on accelerating NMF (NeNMF) from the perspective of convergence and stability, and the resulting optimal NMF simultaneously overcomes the problems of slow convergence, numerical instability, and non-convergence in large-scale data factorization; the method uses the Nesterov gradient to alternatively optimize one factor with another fixed with the Lipschitz constant conducted as step size. The method demonstrates superiority in terms of efficiency and approximation accuracy.

In summary, linear and low-order methods exhibit a wide applicability in *BID* analysis tasks: feature extraction, artifact removal, and EEG classification. Some advanced methods extended from SCA and MF hold potential in big *BID* applications.

4.2 Methods: Linear and High Order

Linear and high-order. DimRed methods directly map the high-order observations (tensor) to the latent variables in a linear manner rather than having to matricize the data as low-order counterparts do (depicted in Figure 5). Methods in this category are generally developed on the top of multi-linear correlation-based analysis and factorization models.

4.2.1 Mathematical Models. Multi-linear correlation-based analysis methods extend the standard counterparts with multi-linear algebra and aim at determining the projection bases in multiple domains to capture the most informative representations presented in original data. MPCA, MLDA, MICA, and TCCA are the most salient instances in this category.

- MPCA seeks for multi-linear projections with most of the variations in each domain preserved the same in PCA. The optimization objective of MPCA is detailed in Appendix A.5, where Equation (A.5.1) can be solved by an alternating projection method [102]. Then MPCA with nonnegativity and sparseness constraints have been developed for superior performance in learning the nonnegative and sparse representations [79, 117].

- MLDA determines multiple interrelated projections under supervision, which maximizes the inter-class scatter while minimizing the intra-class scatter at the same time. The optimization

problem of MLDA is defined in Appendix A.6, which can be solved by the n -mode optimization approach [158]. Then sparse constraints are embedded in MLDA to acquire better discriminant subspace [80], where L_1 and L_2 norms are added to Equation (A.6.1) with regularization parameters α_n, β_{j_n} , such that $\alpha_n \|U_n\|^2 + \sum_{j_n=1}^{J_n} \beta_{j_n} |u_{j_n}|$.

- MICA learns the interactions of multiple sources embedded in N -order data by separately encoding high-order statistics of each source. Given observations $\{X_i \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}\}_{i=1}^m$ and parameters $\{J_n (J_n < I_n)\}_{n=1}^N$, MICA aims to decompose each X_i ($1 \leq i \leq m$) into the mode- n product of N matrices $\{A_n \in \mathbb{R}^{J_n \times I_n}\}_{n=1}^N$ and corresponding core tensor $S_i \in \mathbb{R}^{J_1 \times J_2 \times \dots \times J_N}$, such as $X_i = \{S_i \times_n A_n\}_{n=1}^N$. $\{A_n\}_{n=1}^N$ are derived by the standard ICA algorithm to find statistically independent coefficients on mode- n of $\{X_i\}_{i=1}^m$. Then the **alternating least squares (ALS)** algorithm may apply to find the final solution by minimizing the error: $e = \|\sum_{i=1}^m (X_i - \{S_i \times_n A_n\}_{n=1}^N)\|_F^2$ [147].

- TCCA aims to make the correlation of multiple domains between observed variables from two or more views as maximum as possible. The mathematical definition of TCCA is available in Appendix A.7. The solution to Equation (A.7.1) is obtained by alternating the SVD method in each mode [71]. In addition, a sparse solution to TCCA has been derived using regularized low-rank matrix approximation [152], where $\{U_n\}_{n=1}^N$ and $\{V_n\}_{n=1}^N$ have a small number of nonzero elements.

As described in Section 4.1, TF extends MF to handle high-order observations and explore complicated interdependencies of data elements among multiple domains. Methods in this category include TF and CMTF.

- In TF, **PARAllel FACTor analysis (PARAFAC)**, also known as canonical polyadic decomposition, and the Tucker model [28] are most commonly discussed: (1) PARAFAC excels in a solution's uniqueness under mild conditions, and (2) the core matrix in the Tucker model can be dense while no core associated with PARAFAC makes it much easier to interpret the mechanisms under observations [22].

This survey emphasizes the PARAFAC model for the merits. PARAFAC proposes to decompose multi-way data as a tensor-matrix product of an identity tensor and factor matrices with mathematical definition depicted in Appendix A.8. Traditionally, Equation (A.8.1) may be solved by ALS or its variants. More specifically, the results obtained with ALS can be described as follows:

$$U^{(n)} = X_{(n)} \underset{n' \neq n}{\odot} U^{(n')} \left(\underset{n' \neq n}{\otimes} \left(U^{(n')T} U^{(n')} \right) \right)^{-1}, \quad (2)$$

where $X_{(n)}$ is mode- n matricization of tensor X with size $I_n \times (\prod_{n' \neq n} I_{n'})$, and operators $\odot, \otimes$ respectively denote the Khatri-Rao product and the Hadamard product [28]. Generally, nonnegative constraints are added to capture strongly interpretable factors (nonnegative TF, NTF), where all elements in $\{U^{(n)}\}_{n=1}^N$ are equal to or greater than 0.

In addition, Heiler and Schnörr [53] propose a sparse NTF to enhance its robustness against noises, where sparseness constraint is measured with parameters $s_n^{\min}, s_n^{\max}$ as $s_n^{\min} \leq \frac{1}{\sqrt{I_n-1}} (\sqrt{I_n} - \frac{\|U_n\|_1}{\|U_n\|_2}) \leq s_n^{\max}$. Cichocki et al. [27] propose a more flexible framework using α and β divergence to boost NTF's robustness and the convergence rate. NTF may be further improved by introducing additional constraints, such as smoothness and continuity, for a global convergence and unique solution, and research along this direction remains active.

- CMTF can simultaneously decompose the matrix and tensor in a coupled manner, making it particularly suitable for the data fusion problem. For simplification, coupled tensor and tensor factorization is also defined in the category of CMTF. The optimization objective of CMTF is given in Appendix A.9. Acar et al. [4] use a gradient-based optimization algorithm to compute the factor matrices for a solution to Equation (A.9.1). The result of gradients is obtained as follows:

Table 2. Summary of Linear and High-Order Methods from the Perspective of the Objectives for Optimization

Methods	Learning Pattern	Goal	Objective Function
MPCA	Unsupervised	Maximize variance	the $\arg \max_{U_n^T U_n = I_{J_n}} \sum_{i=1}^m \ \{\mathcal{X}_i \times_n U_n\}_{n=1}^N - \frac{1}{m} \sum_{i=1}^m \{\mathcal{X}_i \times_n U_n\}_{n=1}^N\ _F^2$
MLDA	Supervised	Maximize inter-class divergence and minimize intra-class divergence	$\arg \max_{U_n^T U_n = I_{J_n}} \frac{\sum_{c=1}^C n_c \ \{\hat{\mathcal{X}}_c \times_n U_n\}_{n=1}^N - (\hat{\mathcal{X}} \times_n U_n)_{n=1}^N\ _F^2}{\sum_{i=1}^m \ \{\mathcal{X}_i \times_n U_n\}_{n=1}^N - (\hat{\mathcal{X}}_{c_i} \times_n U_n)_{n=1}^N\ _F^2}$
MICA	Unsupervised	Minimize mean square error (statistically independent sources)	$\arg \min_{\substack{A_n \in \mathbb{R}^{I_n \times J_n} \\ S_i \in \mathbb{R}^{J_1 \times J_2 \times \dots \times J_N}}} \ \sum_{i=1}^m (\mathcal{X}_i - \{S_i \times_n A_n\}_{n=1}^N)\ _F^2$
TCCA	Unsupervised	Maximize correlation coefficient	$\arg \max_{\substack{U_n \in \mathbb{R}^{I_n \times J_n} \\ V_n \in \mathbb{R}^{K_n \times J_n}}} \sum_{i=1}^m \langle \{\mathcal{X}_i \times_n U_n\}_{n=1}^N, \{\mathcal{Y}_i \times_n V_n\}_{n=1}^N \rangle$
PARAFAC	Unsupervised	Minimize mean square error	$\arg \min_{U_n \in \mathbb{R}^{I_n \times R}} \ \mathcal{X} - \{I \times_n U_n\}_{n=1}^N\ _F^2$
CMTF	Unsupervised	Minimize mean square error	$\arg \min_{\substack{U_n \in \mathbb{R}^{I_n \times R} \\ V \in \mathbb{R}^{J \times R}}} \ \mathcal{X} - \{I \times_n U_n\}_{n=1}^N\ _F^2 + \ Y - U^{(m)} V^T\ _F^2$

$$\begin{aligned} \nabla_{U^{(n)}} f &= U^{(n)} \underset{n' \neq n}{\circledast} \left(U^{(n')T} U^{(n')} \right) - X_{(n)} \underset{n' \neq n}{\odot} U^{(n')} + \delta \left(U^{(n)} V^T V - YV \right) \\ \nabla_V f &= V U^{(m)T} U^{(m)} - Y^T U^{(m)}, \end{aligned} \quad (3)$$

where $\delta = \begin{cases} 1 & n=m \\ 0 & n \neq m \end{cases}$. Acar et al. [2] have further developed a structure-revealing CMTF by adding 1-norm penalty, and thus it enables the ability to identify shared and unshared factors [2].

Table 2 offers a comprehensive summary of these methods from the perspective of the objectives for optimization.

4.2.2 Typical Applications. Multi-linear correlation-based analysis and factorization methods are capable of directly handling high-order *BID* with interactions of data elements among multiple domains preserved.

- Multi-linear correlation-based analysis methods excel in classification-oriented tasks based on feature extraction of *BID*, such as SSVEP recognition [166], disease diagnosis [78, 148], ERP classification [54], and fMRI analysis [10]. MPCA can extract structural features correlating to multiple domains from MRI [78], and thus the latent features are used for discriminating **Alzheimer's disease (AD)** subjects from healthy controls with a high classification accuracy than those of linear counterparts. Similarly, Beckmann and Smith [10] utilize MICA to extract features of multi-subject fMRI in spatial, temporal, and subject/session domains, and the features are highly interpretable. Zhang et al. [166] apply TCCA to maximize the overall correlation among canonical variables and find the optimal source components in EEG; it makes SSVEP recognition faster with a higher accuracy than using CCA.

MLDA directly applies in classification tasks for feature extraction in an implicit manner. For instance, Higashi et al. [54] use MLDA with subspace constraints to avoid overfitting in BCI applications, where the parameter spaces are reduced by incorporating the indices of brain functional connectivity; MLDA was highly effective in single-trial ERP classifications. MLDA also applies in

feature extraction of ECG with median filter to remove artifacts and Gabor wavelet transformation to extract P , T waves and QRS complex [148]; MLDA can minimize the signal to mean square error in feature extraction.

- NTF manifests its superiority in capturing multi-domain features of BID with strong interpretability in various scenarios, such as MI/ERP classification [30, 31, 86] and disease diagnosis [82, 103]. Lee et al. [86] determine the discriminative spectral features of MI-EEG using NTF and find the hidden structures from time and class domains; NTF performs well in online MI classification. Analogously, NTF also helps in extracting multi-domain features of visual mismatch negativity (vMMN, a small ERP) with key components regarding ERP context captured using a DIFFIT criterion [30, 31]. Latchoumane et al. [82] and Ma et al. [103] propose a spatio-temporal NTF in analyzing noisy fMRI with the temporal characteristics modeled using the volumetric extraction approach; the overall method boosts the performance of diagnosis of ADHD and AD. NTF is well suited to other neuroengineering tasks of sleep stage identification [75], seizure localization [38], and EEG artifact removal [163].

- CMTF emerges for joint analysis of multi-modal BID , especially those fusing fMRI and EEG. Hunyadi et al. [60] develop a general CMTF framework to capture the most important common patterns, which achieves success in epilepsy analysis. Acar et al. [3] use CMTF to analyze EEG-fMRI collected from patients with schizophrenia; meaningful temporal and spatial signatures can be extracted from the 2-modal BID to differentiate the patients and controls. CMTF may help in a photic driving effect study via EEG-EMG analysis [110].

4.2.3 Potential in Big BID Applications. Numerous attempts have been conducted to boost the factorization methods with the capability of storing and processing big BID . The most salient work along this direction mainly concerns TF and CMTF.

- In TF, it is well known that Equation (2) is computationally demanding along with a large workload incurred by matrix multiplication between $X_{(n)}$ and $\odot_{n' \neq n} U^{(n')}$. How to optimize its solution has long been an open issue with numerous attempts made following three tentative ideas: (1) optimization of core TF algorithms [124, 168], (2) parallelism enhancement of TF [93, 120, 122], and (3) propelling the factorization process with high-performance computing technologies [21, 22, 66, 131] as follows.

Optimization of core algorithms. Efforts have been made to directly optimize the matrix multiplication between $X_{(n)}$ and $\odot_{n' \neq n} U^{(n')}$ as the bottleneck of TF. Low-rank approximation on NTF (lraNTF) is introduced to reduce the time and space complexities [168]; with less rank derived in the algorithm, lraNTF can significantly improve the efficiency and suppress bipolar noises, especially for large-scale problems. Phan et al. [124] propose Fast-ALS to accelerate NTF along with memory requirements reduced; by leveraging a small Khatri-Rao product to compute Equation (2) and avoiding rearranging entries of tensor as much as possible, the method accelerates conventional ALS by 20 to 30 times and even more for tensor with higher orders and ranks.

Parallelism enhancement. Inspired by an idea in the work of Mackey et al. [105], Phan and Cichocki [122] propose Grid-PARAFAC (the core algorithm extended from ALS) to tackle large-scale factorization problems with tremendous-overhead Khatri-Rao products transformed into multiple Hadamard products of relatively small matrices (parallel operations); Grid-PARAFAC can be conducted in a highly parallel manner and process extremely large-scale BID with billions of entries. Papalexakis et al. [120] propose a parallelizable TF method (ParCube) to speed up large-scale sparse factorization with data elements selected by a random sampling technique; ParCube is capable of efficiently conducting sparse factorization and scaling to large datasets. Choi and Vishwanathan [23] present distributed TF (DFacTo) to speed up ALS and the gradient descent technique using sparse matrix-vector products; DFacTo manifests scalability and is 4 to 10 times faster than

counterpart algorithms. In addition, the constrained TF framework is proposed based on alternating direction method of multipliers (ADMoM) [93] with high-overhead computations simplified into distributed and parallel operations, thus opening the door for emerging big data-enabled applications.

Propelling the factorization process with high-performance computing technologies. Scalable and distributed TF (GigaTensor) is proposed to handle large-scale tensor with billions of sizes [66], where the intermediate data explosion problem is avoided by exploiting data's sparseness and carefully redesigning an algorithm on MapReduce; GigaTensor can solve $100\times$ larger tensors than previous counterparts and scale near linearly to the number of machines. Shin et al. [131] propose two distributed TF methods (CDTF, SALS) on MapReduce to decompose large-scale tensor with CDTF updating one parameter at a time while SALS generalizes on the number of parameters updated at a time; both methods (CDTF, SALS) are capable of factorizing 5-order tensor with 1 billion observable entries, 10M mode length, 1K rank, as well as requiring several orders of magnitude less memory. In addition, the hierarchical parallel processing framework over GPU cluster (H-PARAFAC) is developed to enable scalable factorization of large-scale tensor [21]; with a "divide-and-conquer" theory that incorporates coarse-grained model for coordinating the processing of sub-tensors and a fine-grained parallel model for computing each sub-tensor and fusing sub-factors, H-PARAFAC achieves high performance in terms of scalability and efficiency with runtime increasing in the order of n^2 when data volume increases in the order of n^3 . After that, on the basis of H-PARAFAC, incremental PARAFAC (I-PARAFAC) is developed for the factorization of augmenting big *BID* [22], which first searches area encapsulating target factors in each sub-factor space, then locates the target with a fine resolution, and previous factors form the basis of new search with increment considered; I-PARAFAC can factorize huge tensor (550 TB) with runtime in the order of $\frac{1}{n}$ to the number of compute nodes and extra overhead only proportional to the increment.

- Since CMTF focuses on discovering common patterns of multi-modal *BID*, it is more challenging to design a scalable and efficient framework on its top. Few immature works for this purpose exist. Turbo-SMT accelerates CMTF for fMRI analysis to find meaningful features revealing brain activities using biased sampling trick to reduce data volume [119]; Turbo-SMT boosts efficiency by up to $200\times$ with acceptable errors incurred. Beutel et al. [12] develop a flexible CMTF method (FlexiFaCT) on Hadoop using stochastic gradient descent; FlexiFaCT can scale to data with a billion elements under flexible constraints. Scalable CMTF (SCouT) on the distributed MapReduce platform has also been introduced for large-scale matrix-tensor factorization problems [62]; by reorganizing floating point operations and reusing intermediate data, SCouT manifests a scalability linear to data order.

In summary, linear and high-order methods exhibit superiority in processing inherently high-order *BID*, which preserve the structural information defined by the interdependencies among the various modes. A large amount of work is devoted to optimizing the factorization methods toward big *BID* applications, especially TF.

4.3 Methods: Non-Linear and Low Order

Non-linear and low-order. *DimRed* methods capture low-dimensional representations by means of a non-linear transformation/model with observations organized in low order (matrix) (depicted in Figure 6). Methods in this category can be rooted from kernel methods on correlation-based analysis and manifold learning.

4.3.1 Mathematical Models. Kernel methods on correlation-based analysis are non-linear extensions of standard counterparts that apply kernel tricks to turn a linear model into the

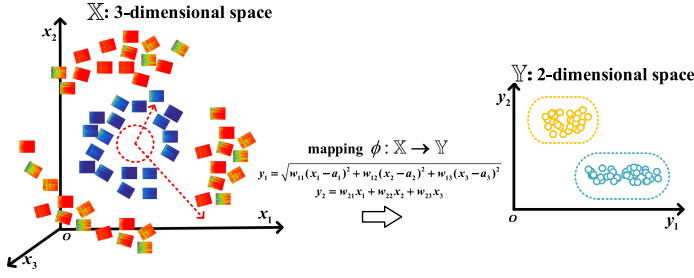


Fig. 6. Illustration of non-linear and low-order *DimRed* methods. Observations (EEG) in 3-dimensional space $\mathbb{X}$ are transformed into those in 2-dimensional space $\mathbb{Y}$ by mapping ϕ : $y_1 = \sqrt{w_{11}(x_1 - a_1)^2 + w_{12}(x_2 - a_2)^2 + w_{13}(x_3 - a_3)^2}$, $y_2 = w_{21}x_1 + w_{22}x_2 + w_{23}x_3$.

non-linear one. Generally speaking, given observations $\{\mathbf{x}_i \in \mathbb{R}^d\}_{i=1}^m$, the basic principle is (1) to map data points into feature space $\mathbb{F}$ via non-linear mapping $\phi: \mathbb{R} \rightarrow \mathbb{F}$, $\mathbf{x}_i \rightarrow \phi(\mathbf{x}_i)$, then (2) to perform SCA in $\mathbb{F}$, where inner product operation of pairs of data points $\langle \mathbf{x}_i, \mathbf{x}_j \rangle$ is changed into $K_{ij} = \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle$. The time-consuming inner product in feature space can be conducted efficiently using kernel function $k(\mathbf{x}_i, \mathbf{x}_j)$ [160], such as multinomial function ($k(\mathbf{x}_i, \mathbf{x}_j) = (\gamma \mathbf{x}_i^T \mathbf{x}_j + r)^d$, $\gamma > 0$) and Gaussian function ($k(\mathbf{x}_i, \mathbf{x}_j) = e^{-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2}$, $\gamma > 0$). Thereby, non-linear properties are naturally embedded in the correlation-based analysis methods, and typical examples include KPCA, KLDA, KCSP, and KCCA.

Manifold learning methods propose to incorporate local or global geometric properties of observations to find basis transformations between high-dimensional space and low-dimensional manifold space with inherent non-linearity. Considering problem solving in *BID* analysis, this survey mainly discusses manifold learning variants of MDS, Isomap, LLE, LPP, and SNE, which differ from each other in definition of the relationship between data points in low-dimensional manifold space.

- MDS is classical manifold learning method that attempts to find low-dimensional embedding space with distances of each pair of data points equally preserved between high-dimensional space and the embedding manifold [63]. Generally, MDS aims to address the optimization problem as defined in Appendix A.10. Then Equation (A.10.1) can be solved via eigenvalue decomposition [63], where the vector norm is computed based on Euclidean distance.

- Isomap extends MDS by incorporating geodesic distances defined in a weighted graph, where the geodesic distance between each pair of data points in high-dimensional space is defined as the sum of edge weights along the shortest path (measured by the Dijkstra algorithm). Isomap basically operates in the three steps conforming to the objective function defined in Equation (A.10.1): (1) to determine the neighbors of each data point and establish the neighborhood graph, (2) to compute the shortest path between each pair of points, and (3) to derive low-dimensional embedding manifold by MDS [139].

- LLE starts with seeking the nearest neighbors of each data point, then employs the linear combination of its neighbors to represent the data point itself. LLE tends to preserve the linear relationship between the target point and its neighbors in the low-dimensional embedding manifold. The optimization objective of LLE is defined in Appendix A.11. Equation (A.11.1) can be solved in the similar way to solve the common sparse eigenvalue problem [127]. Alternatively, Hessian eigenmaps (also known as Hessian LLE) imposes the Hessian estimator in Equation (A.11.1) for optimal results of a non-convex problem [39].

Table 3. Summary of Manifold Learning Methods from the Perspective of the Objectives for Optimization

Methods	Goal	Objective Function
MDS	Keep distance of each pair of data points equal	$\min_{\{y_i\}_{i=1}^m} \sum_{i < j} (\ y_i - y_j\ - \delta_{i,j})^2$
Isomap	Keep distance of each pair of data points equal (geodesic distances)	$\min_{\{y_i\}_{i=1}^m} \sum_{i < j} (\ y_i - y_j\ _{\text{Geo}} - \delta_{i,j})^2$
LLE	Preserve the linear relationship between each point and its neighbors	$\min_{\sum_{j \in Q_i} w_{ij}=1} \sum_{i=1}^m \ x_i - \sum_{j \in Q_i} x_j w_{ij}\ _2^2$ & $\min_{\{y_i\}_{i=1}^m} \sum_{i=1}^m \ y_i - \sum_{j \in Q_i} y_j \hat{w}_{ij}\ _2^2$
LPP	Preserve the neighborhood structure in each point	$\min_{W \in \mathbb{R}^{d \times r}} \frac{1}{2} \sum_{i,j} \ W^T x_i - W^T x_j\ _2^2 s_{ij}$
SNE	Keep the distribution of data points similar	$\min_{i,j} p_{j i} \log \frac{p_{j i}}{q_{j i}}$

• LPP aims to preserve the neighborhood structure in data points and is usually applied to solve the variational problem [52]. The definition of LPP is available in Appendix A.12. Although LPP uses a linear model (Laplacian eigenmaps) for *DimRed*, it excels in keeping the non-linear structures embedded in the original high-dimensional observations.

• SNE considers the correlation between data points from the perspective of probability distribution, which differs from the preceding methods relying on global or local distance metrics. SNE assumes data distribution in high-dimensional space similar to that in the embedding manifold [55]. The optimization objective of SNE can be obtained in Appendix A.13. Equation (A.13.2) can be solved using gradient descent [55]. However, a “crowding problem” may emerge during optimization. Maaten and Hinton [104] then propose a *t*-distributed SNE (*t*-SNE) combining the symmetric SNE and *t*-distribution; here, conditional probabilities $p_{j|i}, q_{j|i}$ are replaced by joint probability $p_{i,j}, q_{i,j}$, and Gaussian distribution is replaced by *t*-distribution. Equation (A.13.2) is then transformed to the alternative as follows with the previous problem avoided:

$$\min \sum_{i,j} p_{i,j} \log \frac{p_{i,j}}{q_{i,j}} \left(q_{i,j} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_l - y_k\|^2)^{-1}} \right), \quad (4)$$

where $p_{i,i} = q_{i,i} = 0$ and $p_{j,i} = p_{i,j}, q_{i,j} = q_{j,i}$.

Table 3 offers a comprehensive summary of manifold learning methods from the perspective of the objectives for optimization (all of the unsupervised learning pattern).

4.3.2 Typical Applications. Kernel methods on correlation-based analysis and manifold learning methods are natural candidates to enable non-linear *DimRed* of *BID* linearly inseparable.

• KPCA is a salient kernel method superior in many neuroengineering tasks, such as P300 detection [144], brain disease diagnosis [100, 132], and EEG/ECG analysis [96, 156]. Turnip et al. [144] enable real-time feature extraction of event-related stimulus EEG benefiting from KPCA’s capability of non-linear adaption, which converges quickly in P300 detection and holds potential in real-time clinical applications. López et al. [100] and Sidhu et al. [132] utilize KPCA to extract features of fMRI from AD and ADHD patients; KPCA can assist clinicians in AD/ADHD diagnosis and outperforms PCA in feature-based applications for brain disease analysis. In addition, KPCA with radial basis kernel can characterize the dynamic complexity of ECG/EEG and construct non-linear

features from there [96, 156]; features revealed by KPCA carry differences statistically significant among various-level mental states ($p < 0.005$), which foster many successful applications, such as sleep apnea detection and mental fatigue estimation.

- A few applications have also been developed with alternative kernel methods, such as EEG classification by KCSP [35, 109], emotion recognition by KLDA [97, 98], and fMRI analysis by KCCA [17]. To tackle the lack of generalization in EEG classification, Dai et al. [35] and Nasihatkon et al. [109] propose a transfer KCSP with the domain-invariant kernel learned via matching distributions between source subjects and target subjects; discriminative information from non-target subjects can then be extracted to enable classification. KLDA has been used to extract non-linear features from spectral powers of single-trial EEG [97, 98], and promising results for emotion recognition have been obtained with accuracy up to 78.49% (valence) and 81.93% (arousal). Blaschko et al. [17] propose a semi-supervised KCCA with Laplacian regularization targeting on small-scale fMRI [17]; by searching for highly correlated directions of fMRI data, KCCA can offer more robust estimate of correlated subspaces and infer significant brain regions in different types of visual processing tasks than conventional KCCA does.

- Manifold learning methods recently have emerged as a powerful tool in *BID* analysis:

- (1) Isomap supports MI classification [90] and neurophysiology assessment [74]. Isomap can effectively assess neurophysiological changes occurring at different anesthetic levels and holds potential in the development of anesthesia monitoring systems [74]. Li et al. [90] propose a supervised Isomap to obtain non-linear manifold structure of MI EEG with geodesic distance matrix calculated based on labeling [90]; the derived features support MI classification with high accuracy.

- (2) LLE has attracted attention in seizure detection [116] and fMRI analysis [129]. Pan et al. [116] propose a weighted LLE to obtain compact representations with internal characteristics in epileptic EEG, which suffices in identifying seizure onset. Shen et al. [129] exploit LLE to capture the underlying disorder-related patterns of fMRI from schizophrenic patients, and thus features work well in discriminating patients from healthy subjects and provide evidence on the important role that certain functional connectivity plays in explaining schizophrenia symptoms.

- (3) LPP has been used to extract features of EEG with samples in the same class close to each other in low-dimensional embedding, improving the performance of EEG classification [155]. Tian et al. [142] improve the neighborhood pixel-based LPP to preserve the most informative representations of fMRI; the method demonstrates effectiveness in semantic feature extraction of fMRI using the Otsu weighted sum of histogram.

- (4) SNE mainly serves the need for seizure detection [15] and MI classification [89]. *t*-SNE captures the non-linear relationship among data elements in epileptic EEG [15]; it proves efficient in seizure detection with the capability of EEG visualization (up to 3D). In addition, a parametric *t*-SNE has been developed for out-of-sample MI classification, and the features manifest obvious clustering distribution with excellent performance achieved [89].

4.3.3 Potential in Big BID Applications. In the big data era, both kernel methods and manifold learning methods are critically challenged because the time and space complexities of their core algorithms are high. There merely exist a few attempts for optimizing KPCA. Manifold learning methods are routinely compute intensive, and how to enable incremental learning remains an urgent issue.

A fast KPCA algorithm with gain adaptation strategy (SMD-PCA) is developed to accelerate KPCA with reciprocal of current estimated eigenvalues incorporated as part of gain vector [49]; the method converges faster and scale better than conventional counterparts do. Choi and Lee [25] propose a dynamic KPCA (DKPCA) to describe the dynamic characteristics of time-lagged data

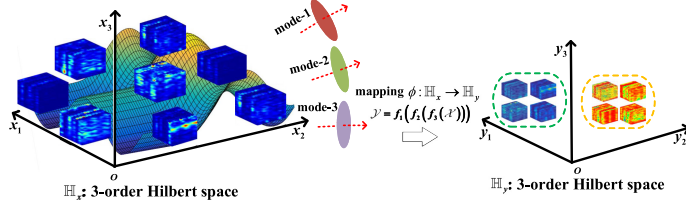


Fig. 7. Illustration of non-linear and high-order *DimRed* methods. The 3-order observations (EEG) in Hilbert space $\mathbb{H}_x$ are transformed into those with comparatively small scale in each domain in another Hilbert space $\mathbb{H}_y$ by mapping $\phi: \mathcal{Y} = f_1(f_2(f_3(X)))$ (f_1, f_2, f_3 denote non-linear functions in three domains).

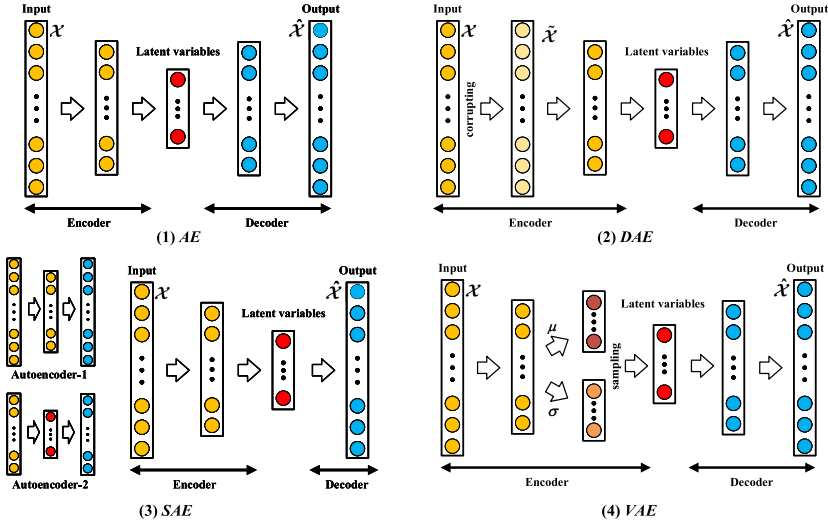


Fig. 8. Neural network architectures of autoencoders.

with the corresponding non-linear properties highlighted; the design aims to monitor an arbitrary process with severe non-linearity, and the resulting error rate and latency both are low.

In summary, non-linear and low-order methods can well match the non-linear characteristics of *BID* and capture corresponding non-linear patterns/structure embedded in *BID*. Nevertheless, efficient and scalable variants on kernel methods on correlation-based analysis and manifold learning methods are rare, which remains a research challenge to adapt them to large-scale *BID* problems.

4.4 Methods: Non-Linear and High Order

Non-linear and high-order. *DimRed* methods capture low-dimensional representations of high-order tensor in a non-linear manner (depicted in Figure 7). Deep learning methods are designed for non-linear learning by nature and can be easily adapted to multi-way/mode operations, and autoencoders are salient in this category.

4.4.1 Mathematical Models. Autoencoders are intrinsically capable of handling high-order data, such as 3D convolution operation, obtaining the compressed representations by means of non-linear transformation such as tanh-function and sigmoid-function. The family of autoencoders consists of raw AE and the variants of DAE, SAE, and VAE. Figure 8 overviews the neural network architectures of autoencoder variants.

- AE is organized as a feedforward neural network consisting of an input layer, an output layer, and multiple hidden layers, where the input and output layers have the same number of nodes trained in a unsupervised manner [56]. The hidden layers are split into an encoder and a decoder, where encoder maps the input (tensor of observed variables) into latent variables and decoder maps the latent variables to the output. Given input $\mathcal{X}$, AE reconstructs $\mathcal{X}$ by minimizing the loss between $\mathcal{X}$ and the output $\hat{\mathcal{X}}$: $\min_{\hat{\mathcal{X}}} D(\mathcal{X}, \hat{\mathcal{X}})$, where $D(\cdot, \cdot)$ denotes the distance between two variables. To find the optimal solution, AE builds two transformations: $\phi : \mathcal{X} \rightarrow \mathcal{L}$ in encoder and $\psi : \mathcal{L} \rightarrow \hat{\mathcal{X}}$ in decoder, and thus the loss is transformed as $\phi, \psi = \arg \min_{\phi, \psi} \|\mathcal{X} - \psi \circ \phi(\mathcal{X})\|_F^2$,

where the Frobenius norm measures the loss. Transformations ϕ, ψ are implicit and determined by parameters w, b of the neural network, which can be solved via classical optimization algorithms, such as stochastic gradient descent (SGD) and the momentum method.

- To improve autoencoders' robustness and generalization capability, DAE partially corrupts the original input by noises and feeds the corrupted input to train the autoencoder [150]. In this sense, DAE makes two assumptions: (1) higher-quality representations are relatively stable and robust to corrupted input, and (2) model tends to highlight inherently informative structure in the distribution of observations. In other words, DAE tends to restore the genuine data from the corrupted input (namely denoising). In the training process, $\tilde{\mathcal{X}} \sim p(\tilde{\mathcal{X}}|\mathcal{X})$ substitutes for $\mathcal{X}$ with the corruption pattern $p(\cdot)$. DAE can then be solved in the same way for AE.

- SAE is introduced to further enrich the representations to be learned from the observations, which builds several autoencoders and makes the output of the hidden layer in one autoencoder connected to the input of the corresponding hidden layer in another autoencoder [130]. The learning process consists of three steps: (1) to train one autoencoder fed with known observations, (2) to use the representations obtained from the previous autoencoder as input to train the next autoencoder, and (3) to jointly optimize these autoencoders for fine tuning on completion of training all individual autoencoders. Actually, in the first two steps, SAE quickly fits the observations' structure of a coarse grain; the third step then derives the fine-grained low-dimensional representations.

- In contrast to the preceding, VAE acts as a generative model. VAE specializes in understanding the underlying causal relations embedded in observations, and it generates high-quality data as counterparts [72]. Here the basic principle is to map input into the hidden layer conforming to a certain distribution rather than applying fixed values as in AE, where the distribution is expressed by parameter vectors, such as mean vector and deviation vector in Gaussian distribution.

The latent variables are sampled by parameter vectors for the use of a decoder. Given observations $\mathcal{X}$, the marginal likelihood $p_\theta(\mathcal{X})$ in the context of VAE can be written as $\log p_\theta(\mathcal{X}) = D_{\text{KL}}(q_\phi(\mathcal{Z}|\mathcal{X})\|p_\theta(\mathcal{Z})) + \mathcal{L}(\theta, \phi; \mathcal{X})$, where $D_{\text{KL}}(\cdot\|\cdot)$ is the KL divergence between the approximate distribution and the true posterior distribution, and θ, ϕ are the generative parameters and variational parameters. Since KL divergence is nonnegative, $\mathcal{L}(\theta, \phi; \mathcal{X})$ can be regarded as the lower bound of $\log p_\theta(\mathcal{X})$, which acts as optimization objective of VAE and can be derived by the standard stochastic gradient variational Bayes estimator [72].

4.4.2 Typical Applications. Autoencoders have been widely applied in classification-oriented tasks of *BID* with the merits of adaptive and data-driven learning.

- SAE is superior in various scenarios of *BID* analysis, such as seizure detection [94, 136], emotion recognition [20, 64], and ECG/EEG classification [44, 137]. An unsupervised framework based on SAE has been developed to automatically extract informative features from epileptic EEG for seizure detection [94, 136]; the framework achieves a detection accuracy of 96% and well adapts to seizures of varying patterns. SAE applies in discovering unknown correlation of features in EEG to support emotion recognition [64]; the method can recognize different levels of valence and

arousal with accuracy of 49.52% and 46.03%, respectively. Discrepancy in data distribution always occurs with EEG due to fatigue of operators, difference with electrode placements, and variation in impedances. Chai et al. [20] employ SAE with a domain adaptation strategy to obtain common representations across training and test domains; a reliable emotion recognition has then been enabled with accuracy up to 77.88% (subject-to-subject) and 81.81% (session-to-session).

In addition, the SAE-based framework emerges to simultaneously compress and reconstruct ECG and EEG using a split Bregman technique for a real-time ECG/EEG classification [44, 137]; the framework operates at least an order magnitude faster than the counterparts do. SAE also prevails in P300 detection [146], sleep stage detection [143], and diagnosis of other brain disease [135].

- Other autoencoder variants also have a small number of *BID* applications. A deep convolutional AE framework with 27 hidden layers has been developed for unsupervised feature learning [154, 159]. The framework enables effective analysis of epileptic EEG and supports efficient ECG compression (32.25 times), which holds potential in e-health applications. DAE performs well in handling noisy and non-stationary *BID*. Li et al. [88] utilize DAE to learn features of incomplete EEG corrupted by a diversity of artifacts, such as EMG and background activities; DAE can decode incomplete EEG assisted by adaptive encoding and support reliable MI classification in BCI systems. Qiu et al. [125] develop a DAE-based framework for robust epilepsy analysis, where sparsity constraint applies to the hidden layer for a more concise representations of EEG considering the intensive non-stationarity; the framework supports near-perfect seizure detection.

VAE focuses on applications based on MI/ERP classification [13, 115]. Bi et al. [13] propose a semi-supervised VAE for ERP-based driver-vehicle interfaces; it is then gradually improved by updating the semi-supervised decoder with newly collected unlabeled data, where the overhead of model training can then be significantly reduced; the method manifests good applicability in brain-controlled vehicles. An adversarial VAE model emerges to learn subject-invariant representations of MI with VAE and the adversarial network simultaneously trained [115]; the model is capable of fast subject-transfer learning and paves the way toward semi-supervised transfer learning in BCI applications. A VAE variant maintains five hidden layers in both the encoder and decoder, which is designed to learn visual features from cortical activities recorded in fMRI [50]; the features are useful in predicting video-evoked cortical responses with a high accuracy and hold potential in explaining the phenomenon regarding cortical responses.

4.4.3 Potential in Big *BID* Applications. Autoencoders operate in a data-driven manner and exhibit built-in high parallelisms. These features make them natural candidates for end-to-end automatic operations on large-scale *BID*, such as NVIDIA's GPU-accelerated library.

For instance, online feature learning schemes on AE and DAE are proposed to determine the optimal model for large-scale datasets; it merges similar features to obtain compact representation with the residual function minimized and also prevents the overfitting problem [87, 169]. Recently, autoencoder-based methods have adverted to handle large-scale *BID* over cloud. Hosseini et al. [57, 58] develop a cloud-based framework for seizure prediction and localization, where SAE is exploited to sustain unsupervised feature extraction of EEG/ECoG; the framework can support real-time analysis, automatic computing, and management of big EEG/ECoG data. In addition, a scalable and distributed AE emerges upon Apache Spark (a cluster of GPU nodes over cloud) with efficiency and scalability highlighted, which aims to enable modeling and analytics toward big fMRI data analysis [107].

In summary, non-linear and high-order methods are capable of unsupervised feature learning of *BID* with intensive non-linearity and routinely high order. These methods (autoencoders) also exhibit superiority in processing large-scale *BID* with an end-to-end manner.

Overall, the most representative *DimRed* methods for *BID* are presented with detailed mathematical principles. For each method, its typical applications in small-scale *BID* and potential in big *BID* are emphasized. Table 4 presents a summary of *DimRed* methods in practical applications and optimizations for large-scale *BID*.

5 DISCUSSION

Clearly, dimensionality reduction is at the very core of *BID* analysis. It directly determines the effectiveness, interpretability, efficiency, and scalability in solving neuroscience and engineering problems. This section discusses the advantages and shortcomings of the surveyed methods with comparisons presented. Then it provides insights into the trends of this research area and challenges to be addressed in the near future.

5.1 Summary and Comparison

Existing methods are largely rooted from following mathematical bases/models: correlation-based analysis (SCA, kernel methods on correlation-based analysis, and multi-linear correlation-based analysis), factorization, manifold learning, and autoencoders. For the same problem under examination, methods with different mathematical basis/models may obtain totally different results and significantly vary in applicabilities due to their heterogeneity:

(1) Correlation-based analysis:

- SCA methods prevail with the simplicity of model solving and interpretability, which have manifested almost the most widespread applications for *DimRed* of *BID* over other methods: ICA for artifact removal, CSP variants in EEG-based BCI tasks, and CCA for SSVEP recognition. However, these methods rely on strict assumptions of statistically linear independence of latent variables, such as PCA and ICA, which cannot guarantee effectiveness and robustness when coping with *BID* with intensive noises and outliers, or *BID* are intrinsically unable to match the preceding assumption. SCA methods present poor performance for *DimRed* of highly non-linear and non-stationary *BID* as the core algorithms are conducted via linear mapping.

- Kernel methods (a non-linear extension of SCA) well suit the non-linear *DimRed* tasks where a linear candidate fails to reveal the implicitly non-linear characteristics of *BID*. Typically, KPCA has exhibited superiority in *DimRed* of fMRI data from AD and ADHD patients than PCA [100, 132] due to the capability of obtaining non-linear principal components. However, there exist several issues in comparison with SCA counterparts: (1) if linearity in the *BID* dominates, kernel methods may perform worse due to the overfitting; (2) kernel methods lose the set of weights as the structural information of *BID* like the forms described in SCA weaken the interpretability; and (3) kernel methods are much more computation and data intensive and generally are subject to a performance bottleneck in massive *BID* analysis.

- Multi-linear correlation-based analysis (tensorized SCA) excels in constructing the structural features of high-order *BID* defined by the interdependencies among the various data domains/modes/dimensions. Similar to SCA, strict assumptions are mandatory such as orthogonality in MPCA, making them difficult to converge in model solving and very sensitive to parameter settings. The optimization process of tensorized SCA also incurs excessive overheads as correlating multiple data domains.

(2) Factorization methods:

Factorization methods derive the latent factors (possibly as indicators of the malfunctional brain dynamics) without strict assumptions with the most desirable merits: the models are generally solvable, and it is easy to obtain the intuitive interpretations to the underlying mechanism under observations. TF can handle high-order *BID* possibly originated from multiple sources to extract the structural factors correlating the data domains/modes all together

Table 4. Summary of *DimRed* Methods for *BID* (Applications in Small-scale *BID*; Optimizations for Processing Large-scale *BID*)

Category	Method	Application (Small-Scale <i>BID</i>)	Optimization (Large-Scale <i>BID</i>)
Linear & low order	PCA	Epileptic EEG [5, 133]	FastPCA [128]
		Artifact removal [19, 140]	GS-PCA [7]
		Emotion recognition [64]	disPCA [92]
		Schizophrenia discrimination [18]	sPCA [41]
	LDA	Seizure detection [170]	ILDA [118]
		MI/ERP classification [161, 165]	GSVD-ILDA [167]
	CSP	EEG-based BCI [8, 101, 113, 141, 157]	
	ICA	Artifact removal [61, 149]	CUDA-ICA [126], FPGA-ICA [40]
Linear & high order	CCA	Artifact removal [17, 37]	
		SSVEP recognition [95, 164]	
		Multi-modal <i>BID</i> [32]	
	MF	EEG feature extraction [84, 85]	Online-MF [106], OR-NMF [48]
		fMRI analysis [6, 99]	DSGD-MF [43], DFC-MF [105]
		Artifact removal [36]	NeNMF [47]
	MPCA	Disease diagnosis [78]	
	MLDA	ERP classification [54]	
		Disease diagnosis [148]	
	MICA	fMRI analysis [10]	
	TCCA	SSVEP recognition [166]	
	TF	MI/ERP classification [30, 31, 86]	IraNTF [168], Fast-ALS [124]
		Disease diagnosis [82, 103]	GigaTensor [66], Parcube [120]
		Sleep stage [75]	DFacTo [23], CDTF/SALS [131]
		Seizure localization [38]	Grid-PARAFAC [122], ADMoM [93]
		Artifact removal [163]	H-PARAFAC [21], I-PARAFAC [22]
	CMTF	Epilepsy analysis [60]	Turbo-SMT [119], SCouT [62]
		Schizophrenia study [3]	FlexiFaCT [12]
Non-linear & low order	KPCA	P300 detection [144]	SMD-PCA [49]
		Disease diagnosis [100, 132]	DKPCA [25]
		EEG/ECG analysis [96, 156]	
	KLDA	Demotion recognition [97, 98]	
	KCSP	EEG classification [35, 109]	
	KCCA	fMRI analysis [17]	
	Isomap	MI classification [90]	
		Neurophysiology assessment [74]	
	LLE	Seizure detection [116]	
		fMRI analysis [129]	
Non-linear & high order	LPP	EEG classification [155]	
		fMRI analysis [142]	
	SNE	Seizure detection [15]	
		MI classification [89]	
	AE	EEG/ECG feature learning [154, 159]	
	DAE	MI classification [88]	
		Epileptic EEG analysis [125]	
	SAE	Seizure detection [94, 136]	
		Emotion recognition [20, 64]	
		ECG/EEG classification [44, 137]	Cloud computing solution [57, 58, 87, 107, 169]
		P300 detection [146]	
		Sleep stage [143]	
	VAE	Disease diagnosis [135]	
		MI/ERP classification [13, 115]	
		fMRI analysis [50]	

Table 5. Qualitative Comparison of the *DimRed* Methods in This Survey

Methods	Criteria						
	Effectiveness				Interpretability	Scalability	Efficiency
	Linear	Non-linear	Low order	High order			
SCA	✓		✓		✓	✓	✓
Kernel methods on SCA		✓	✓				
Multi-linear SCA	✓		✓	✓	✓		
Factorization	✓		✓	✓	✓	✓	✓
Manifold learning		✓	✓				
Autoencoders	✓	✓	✓	✓		✓	✓

for an enhanced characterization of physiological conditions. Note that it is an active direction to enhance the scalability and efficiency of factorization methods.

(3) *Manifold learning methods*: Manifold learning methods largely apply to EEG or fMRI to explore the embedded low-dimensional manifold. These methods vary from each other mainly in the relationship defined in the manifold space, such as geodesic distances (Isomap) and linear relationship with neighbors (LLE). Note that these methods are application specific with generality ignored, and there exist comparatively few applications concerning the *DimRed* of BID along this direction. How to enable incremental learning remains an urgent issue to solve the performance bottleneck.

(4) Autoencoders excel in the capability of automatic learning driven by data, as well as the built-in high parallelism. The merits make them particularly suitable for end-to-end automatic operations. Some variants, such as VAE, also hold potential in simulating high-quality *BID* for training and testing purposes. Like other deep learning methods, autoencoders also suffer from a lack of interpretability of the underlying mechanisms. Their effectiveness depends on whether high-quality and large-scale training data are available as determined by the model's data-driven nature.

To sum up, Table 5 presents a qualitative comparison among the *DimRed* methods. Note that three categories are described regarding the correlation-based analysis family.

5.2 Research Challenges

Although grand success has been achieved in this research area, this survey still foresees a numbers of unsolved issues and challenges to be addressed. This study focuses on online *BID* analysis, joint *DimRed* of multi-modal *BID*, advanced TF, *BID* restoration, and metric learning with future research directions illustrated in Figure 6:

(1) **Online BID analysis**: Most *DimRed* methods are still limited to applications of offline *BID* analysis. The recently booming neuroengineering technologies such as BCI and brain e-health demand real-time outputs thus timely decision may be made to sustain the operation of online models/systems. Emerging platforms for big EEG data analysis propel the development of *DimRed* methods of dramatically enhanced scalability and efficiency to support seizure detection and prediction [9, 57, 58]. But more sophisticated tasks such as interpretation of cognitive states are still rare, which demand more advanced computing technologies and incorporation of cutting-edge predictive models to solve the complicated problems together.

(2) **Joint DimRed of multi-modal BID**: The spatio-temporal resolutions of brain imaging techniques significantly differ from each other. EEG and MEG can provide high temporal resolution (milliseconds) and fMRI has a high spatial resolution (full brain), but not vice versa. It is

desirable to complement a type of *BID* with the other *BID* of a heterogeneous modality, thus benefiting the understanding of brain dynamics from the merits of both. Joint *DimRed* of multi-modal *BID* can provide more reliable conclusions via orchestrating different information inherently carried by multi-modal *BID*—for example, epileptic seizures not only cause violent nervous activities (EEG) but also simultaneously result in events of heartbeat pattern (ECG) and breathing (EMG). Research along this direction is still in its infancy: existing works basically perform *DimRed* on each *BID* and simply combine features from multiple modalities with marginal performance gain. Clearly, an ensemble of multi-modal *BID* are more than a sum of several individual *BID* sets. Novel theories and methods for joint *DimRed* and fusion of multi-modal *BID* are expected, possibly rooted from factorization methods and autoencoders, to reveal the hidden mechanisms interlaced by the ensemble. Those hold great potential in functional brain studies, such as brain disease diagnosis, neurophysiological assessment, and treatment intervention.

(3) **Advanced TF:** TF methods have achieved grand success in high-order *BID* analysis with the capability of deriving structural features correlating multiple interacting domains. The methods routinely suffer from excessive overhead and risks of falling in local optimum. Numerous attempts have been made to improve the efficiency, scalability, and robustness of factorization methods, but the performance bottleneck remains an open issue. Moreover, there exists no qualitative or quantitative solution to the local optimum problem. The new method of tensor networks [26, 123] can represent a tensor as sets of sparsely interconnected lower-order tensors to reduce the computing resources, but this attempt is still far from mature. Furthermore, second-order optimization can be an interesting direction to accelerate the convergence of factorization and reduce the risk of local optimum.

(4) **BID restoration:** *BID* are routinely superimposed by a high level of noises and interferences—for example, EEG is prone to contamination by environmental noises, such as EOG and EMG. The derived low-dimensional features will inevitably be distorted. The mainstream approaches to artifact removal largely depend on the empirical analysis with ideal assumptions, such as statistics of frequency bands and noises originated from particular brain regions. Their designs ignore the needs from *DimRed* and are constrained to offline work mode. It becomes extremely difficult to restore the genuine recordings of neuronal activities. Consequently, the effectiveness of the corresponding features cannot be guaranteed, and so the reliability of the front-end system cannot be ensured either. In the big *BID* paradigm, deep learning methods hold great potential in tackling this problem, especially in supporting adaptive artifact removal via data-driven models. As the inverse of *DimRed* (see also Section 5.1), generative models will prevail in simulating “realistic” *BID* such as generative adversarial networks [83] and VAE.

(5) **Metric learning:** Most of the existing *DimRed* methods exploit standard distance metrics to measure the relationship among data points either implicitly or explicitly, such as Euclidean distance in PCA, but how well those pre-set metrics are suited for the *DimRed* of particular *BID* often remains unclear. Metric learning enables automatic construction of the task-specific embedding function from data itself that can enlarge the distances between dissimilar data points while reducing the distances between similar ones [151].

Deep metric learning has recently attracted much attention for providing better solutions via deeper non-linear feature learning, especially the Siamese network model and the Triplet network model [68]. The Siamese network with graph convolutions [76] has been applied in measuring the similarity among brain connectivity networks, then *DimRed* of fMRI data is achieved to support disease diagnosis of high performance. Nevertheless, the applications of metric learning in *BID* are far from immature at present. Researchers and practitioners need to focus on *BID* sampling strategy, metric-based loss function, and network architecture for critical applications of *BID* analysis.

Table 6. Research Challenges and Future Research Directions

Research Challenges	Future Directions
Online <i>BID</i> analysis	◊ Brain healthcare [57, 69] ◊ Online BCI system [70]
Joint <i>DimRed</i> of multi-modal <i>BID</i>	◊ Multi-modal <i>BID</i> fusion [162]
Advanced TF	◊ Tensor network [123] ◊ Second-order optimization [111, 145]
<i>BID</i> restoration	◊ Automatic noise/interference removal [114] ◊ <i>BID</i> augmentation [42, 81]
Metric learning	◊ <i>BID</i> sampling strategy [68] ◊ Definition of metric-based loss [68, 76] ◊ Design of network architecture [76]

6 CONCLUSION

Dimensionality reduction theories and methods manifest unrivaled potential in revealing key insights to *BID* via offering low-dimensional and tiny features. This study is intended to bridge the gap between the diversity of dimensionality reduction methods and the huge diversity of *BID*, and more importantly the neuroscience and engineering tasks.

This survey overviews the most salient work along this direction conforming to a 3-dimensional taxonomy with respect to (1) the scale of *BID*, in which the design with this consideration is important for the potential applications; (2) the order of *BID*, in which a higher order denotes more *BID* attributes manipulatable by the method; and (3) linearity, in which the method's degree of linearity largely determines the "fidelity" in *BID* exploration. Meanwhile, qualitative evaluations on these works are performed in terms of effectiveness, interpretability, efficiency, and scalability.

The key findings from the survey are as follows:

- (1) In the correlation-based analysis family, standard correlation-based analysis methods prevail with the simplicity of model solving and interpretability that have manifested almost the most widespread applications for *DimRed* of *BID* but suffer from the strict assumptions of statistically linear independence of latent variables; kernel methods are well suited for the scenarios where non-linear mapping is a must at the cost of sacrificing the interpretability, efficiency, and scalability; and multi-linear correlation-based analysis excels in constructing the structural features of high-order *BID*, but the optimization process incurs excessive overheads.
- (2) Factorization models are generally solvable and excel in intuitive interpretations to the observations demanding no strict assumptions. TF can handle high-order *BID* possibly originating from multiple sources for an enhanced characterization of physiological conditions. It is an active direction to enhance the scalability and efficiency of factorization methods.
- (3) Manifold learning methods largely apply to EEG or fMRI with various relations defined in the manifold space. These methods are largely application specific, showing advantages in solving problems with intensive non-linearity. How to enable incremental learning remains an urgent issue to solve the performance bottleneck.
- (4) Autoencoders are particularly suitable for end-to-end automatic operations. Autoencoders are data driven by nature and perform well with high-quality training data available. A lack of interpretability is the common weakness of autoencoders.

This study spots out the open technical issues and proposes research challenges in online *BID* analysis, joint *DimRed* of multi-modal *BID*, advanced TF, *BID* restoration, and metric learning. In the near future, research issues centering on big *BID* applications will be the priority in this area of great potential.

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